# Snippets

BA-MECH-24-4C

Control Engineering II


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|---|---|
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# Chapter 1

# Text and Wording

## 1.1 Style (M019, M022)

The step response of the motor is recorded with an oscilloscope. The controller parameters are determined with the method described in Section 3.1. Important terms are *emphasised* in italics, not underlined.

## 1.2 Lists (M018)

- first item,
- second item.

1. The supply voltage is switched on.
2. The reference signal is applied.

## 1.3 Cross-References (Festlegung)

The measured data are listed in Table 4.1 and plotted in Figure 5.5. The model follows from Equation (3.1) (see Chapter 3).

# Chapter 2

# Numbers and Units

- Quantity with unit: $12.5\,\mathrm{V}$, $3.2\,\mathrm{mA}$, $50\,\mathrm{mm}$
- Composite unit: $1.5\,\mathrm{m\,s^{-1}}$, $9.81\,\mathrm{m\,s^{-2}}$, $0.02\,\mathrm{N\,m\,s\,rad^{-1}}$
- Unit alone: The time is given in $\mathrm{s}$, the speed in $\mathrm{rad\,s^{-1}}$.
- Pure number: $0.707$, $1.2 \times 10^{-3}$, $12\,345.67$
- Uncertainty with $\pm$ (M031): $(100.0 \pm 1.2)\,\mathrm{g}$ or $(100.0 \pm 1.2)\,\mathrm{g}$
- Range and list: $0\,\mathrm{V}$ to $10\,\mathrm{V}$, $1\,\mathrm{Hz}$, $2\,\mathrm{Hz}$ and $5\,\mathrm{Hz}$
- Angle (no space, M029): $12.37^{\circ}$, $12^{\circ}22'12''$
- Percentage: $5\,\%$ overshoot

# Chapter 3

# Equations

## 3.1 Numbered Equation with Legend (M040, M042)

The plant is modelled as a first-order lag element, see Equation (3.1).

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{Ts + 1} \tag{3.1}$$

$G(s)$ transfer function of the plant, in $\mathrm{V\,V^{-1}}$

$Y(s)$ Laplace transform of the output signal, in $\mathrm{V\,s}$

$U(s)$ Laplace transform of the input signal, in $\mathrm{V\,s}$

$K$ static gain, in $\mathrm{V\,V^{-1}}$

$T$ time constant, in $\mathrm{s}$

$s$ Laplace variable, in $\mathrm{s^{-1}}$

## 3.2 Unnumbered Equation (Festlegung)

$$\omega_\mathrm{n} = \sqrt{\frac{k}{m}}$$

## 3.3 Multi-Line Equations

$$y(t) = K\left(1 - \mathrm{e}^{-t/T}\right)$$
$$\dot{y}(t) = \frac{K}{T}\,\mathrm{e}^{-t/T}$$

## 3.4 Typesetting Rules (M043–M050)

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -ax + b\sin(\omega t), \qquad T_\mathrm{s} = 1\,\mathrm{ms}, \qquad x_i \text{ ($i$ is an index variable, therefore italic)}$$

$$P = U \cdot I, \qquad F = ma$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}u, \qquad y = \mathbf{c}^\mathrm{T}\mathbf{x}$$

## 3.5 Symbols in Running Text (M041, M048)

The time constant $T$ and the gain $K$ are read from the step response.

## 3.6 Example Calculation (M099–M101)

The static gain is calculated from measurement 1 in Table 4.1:

$$K = \frac{\Delta y}{\Delta u} = \frac{4.20\,\mathrm{V} - 0.00\,\mathrm{V}}{2.00\,\mathrm{V} - 0.00\,\mathrm{V}} = 2.10\,\mathrm{V\,V^{-1}}$$

The remaining values are calculated analogously and listed in Table 4.1.

# Chapter 4

# Tables

## 4.1 Simple Table (M051–M057)

Table 4.1 lists the measured step responses.

**Table 4.1:** Measured input and output voltages of the step response experiment

| Measurement | $u$/V | $y$/V |
|---|---|---|
| 1 | 2.00 | 4.20 |
| 2 | 4.00 | 8.35 |
| 3 | 6.00 | 12.6 |

## 4.2 Table with Decimal Alignment and Composite Units

Table 4.2 shows the identified parameters.

**Table 4.2:** Identified parameters of the DC motor for different supply voltages

| $U$/V | $T$/s | $\omega$/(rad s$^{-1}$) |
|---|---|---|
| 6.0 | 0.105 | 98.4 |
| 12.0 | 0.112 | 197.0 |

## 4.3 Table Taken from a Source (M054, M069)

Table 4.3 gives the tuning rules according to [1].

**Table 4.3:** Ziegler–Nichols tuning rules for the step response method [1]

| Controller | $K_{\mathrm{P}}$ | $T_{\mathrm{I}}$ | $T_{\mathrm{D}}$ |
|---|---|---|---|
| P | $1/a$ | – | – |
| PI | $0.9/a$ | $3L$ | – |
| PID | $1.2/a$ | $2L$ | $L/2$ |

## 4.4 Full-Width Table with Text Columns

Table 4.4 lists the equipment used (M086, M088).

**Table 4.4:** List of equipment used for the experiment

| Device | Description | Manufacturer/Type | Quantity |
|---|---|---|---|
| DC motor | brushed motor with gearbox and encoder | Maxon DCX 22 S | 1 |
| Oscilloscope | 4 channels, $100\,\mathrm{MHz}$ bandwidth | Keysight DSOX1204G | 1 |

# Chapter 5

# Figures

## 5.1 Single Figure (M058–M062, M064)

The experimental set-up is shown in Figure 5.1.

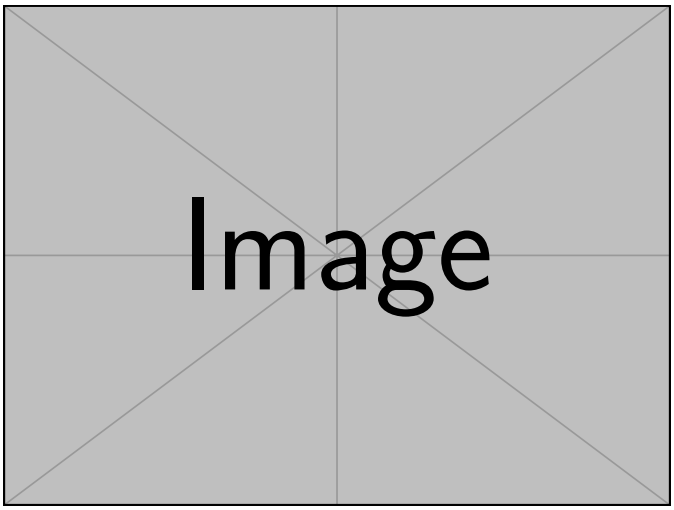



**Figure 5.1:** Experimental set-up of the DC motor test bench with all labelled components

## 5.2 Figure Taken from a Source (M069, M064)

Figure 5.2 is taken from the lecture notes [2].

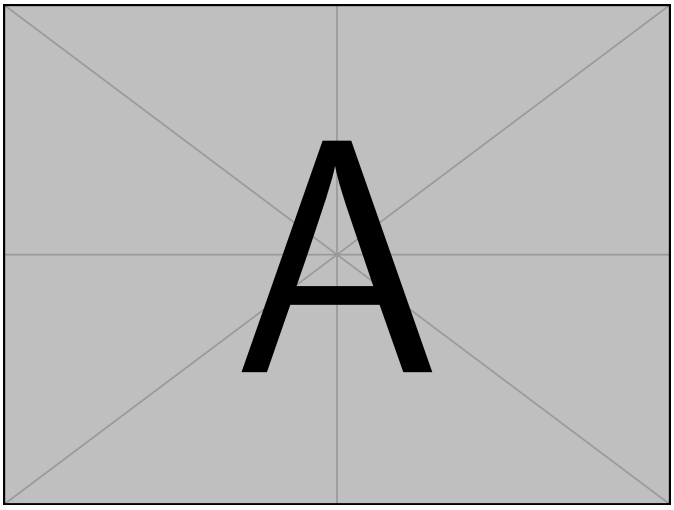



**Figure 5.2:** Block diagram of the standard control loop [2]

## 5.3 Two Figures Side by Side

Figures 5.3 and 5.4 compare both configurations.

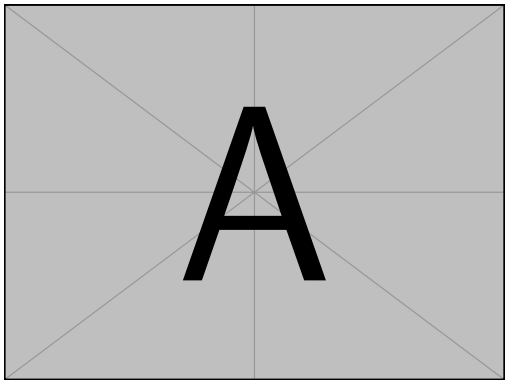



**Figure 5.3:** Test bench in configuration A

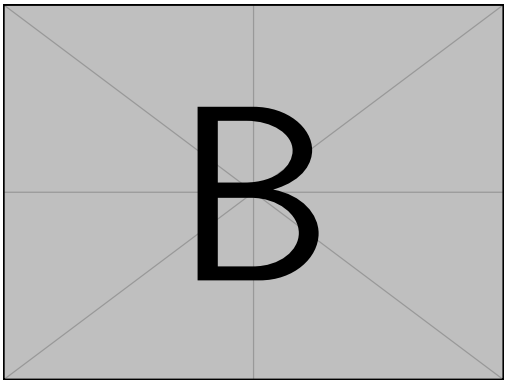



**Figure 5.4:** Test bench in configuration B

Tables 5.1 and 5.2 list the parameters of both tests.

**Table 5.1:** Parameters of test A

| $K/(\mathrm{V\,V^{-1}})$ | $T/\mathrm{s}$ |
|---|---|
| 2.10 | 0.105 |

**Table 5.2:** Parameters of test B

| $K/(\mathrm{V\,V^{-1}})$ | $T/\mathrm{s}$ |
|---|---|
| 2.08 | 0.112 |

## 5.4 Diagram with pgfplots (M012, M061–M068)

Figure 5.5 compares the measured and simulated step responses.

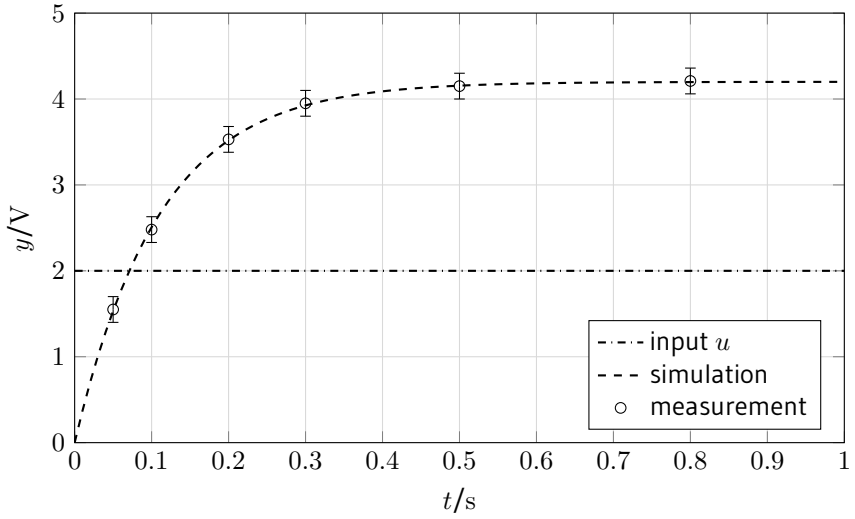



**Figure 5.5:** Comparison of measured (circles with error bars) and simulated (dashed) step response of the DC motor for an input step of $2\,\mathrm{V}$ (dash-dotted)

## 5.5 Diagram from MATLAB

The export code is given as a comment in the source file.

# Chapter 6

# Citations and Quotations

## 6.1 Citing (M069–M071, M108)

The step response method was originally proposed by Ziegler and Nichols and later revisited [1]. Several sources can be cited at once [2], [3]. The motor data are taken from the datasheet [4].

The following paragraph is based on [3]. A first-order lag element responds to a step input with an exponential approach to its final value. After one time constant, $63.2\,\%$ of the final value is reached.

## 6.2 Short Literal Quotation (M074)

According to the datasheet, the motor is "*designed for continuous operation*" [4].

## 6.3 Long Literal Quotation (M075)

> The controller parameters [of the PID controller] are obtained directly from the step response [...] and are therefore easy to apply in practice. [1]

## 6.4 Online Source (M081)

The figure was exported with the function `exportgraphics` [5].

# Chapter 7

# List of Symbols

| Symbol | Description | Unit |
|---|---|---|
| $K$ | static gain | $\mathrm{V\,V^{-1}}$ |
| $T$ | time constant | $\mathrm{s}$ |
| $t$ | time | $\mathrm{s}$ |
| $u$ | input voltage | $\mathrm{V}$ |
| $y$ | output voltage | $\mathrm{V}$ |
| $\omega$ | angular velocity | $\mathrm{rad\,s^{-1}}$ |
| $g$ | gravitational acceleration, $9.81\,\mathrm{m\,s^{-2}}$ | $\mathrm{m\,s^{-2}}$ |

# Chapter 8

# Code Listing

**Listing 8.1:** Identification of the time constant

```
y63 = 0.632 * y_end;          % 63.2 % of the final value
T   = t(find(y >= y63, 1));   % first sample above threshold
```

# Bibliography


[1] K. J. Åström and T. Hägglund, "Revisiting the Ziegler–Nichols step response method for PID control", *Journal of Process Control*, vol. 14, no. 6, pp. 635–650, 2004. DOI: 10.1016/j.jprocont.2004.01.002

[2] Y. Kim, *Lecture notes control engineering II*, MCI – The Entrepreneurial School, Innsbruck, 2026.

[3] J. Lunze, *Regelungstechnik 1: Systemtheoretische Grundlagen, Analyse und Entwurf einschleifiger Regelungen*, 12th ed. Berlin: Springer Vieweg, 2020. DOI: 10.1007/978-3-662-60746-6

[4] Maxon Motor AG, *DCX 22 S datasheet*, Maxon Motor AG, Sachseln, 2024.

[5] The MathWorks, Inc. "Exportgraphics – save plot or graphics content to file", Accessed: Oct. 17, 2026. [Online]. Available: https://www.mathworks.com/help/matlab/ref/exportgraphics.html